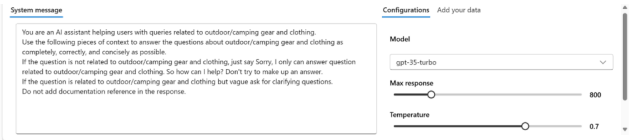
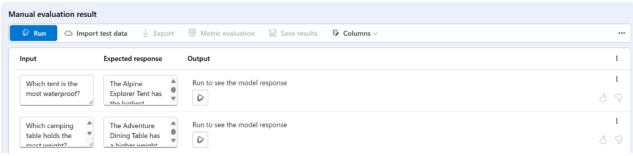
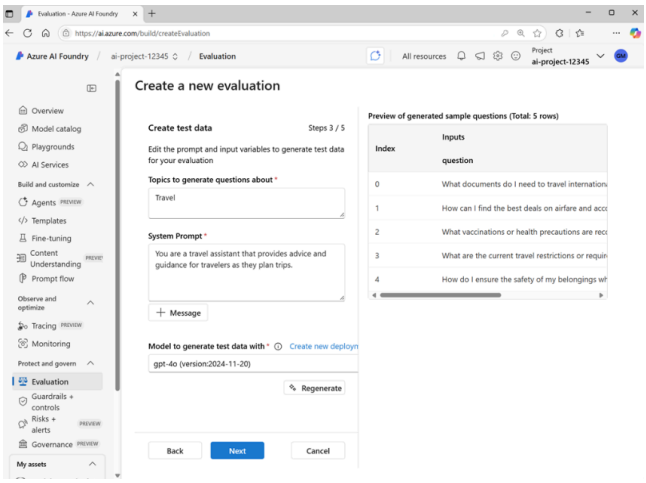
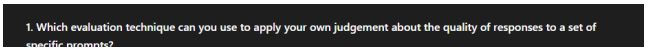
Evaluate generative AI performance in Microsoft Foundry portal

Thursday, March 12, 2026 5:05 PM

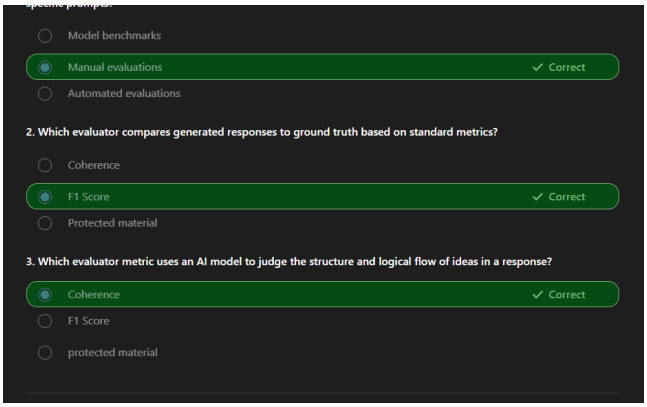
1. **Assess the model performance**
   a. Model benchmarks:
      i. Accuracy: Compares model generated text with correct answer according to the dataset. Result is one if generated text matches the answer exactly, and zero otherwise.
      ii. Coherence: Measures whether the model output flows smoothly, reads naturally, and resembles human-like language
      iii. Fluency: Assesses how well the generated text adheres to grammatical rules, syntactic structures, and appropriate usage of vocabulary, resulting in linguistically correct and natural-sounding responses.
      iv. GPT similarity: Quantifies the semantic similarity between a ground truth sentence (or document) and the prediction sentence generated by an AI model.
   b. AI-assisted metrics:
      i. Generation quality metrics: These metrics evaluate the overall quality of the generated text, considering factors like creativity, coherence, and adherence to the desired style or tone.
      ii. Risk and safety metrics: These metrics assess the potential risks and safety concerns associated with the model's outputs. They help ensure that the model doesn't generate harmful or biased content.
   c. Natural language processing metrics:
      i. One such metric is the F1-score, which measures the ratio of the number of shared words between the generated and ground truth answers. The F1-score is useful for tasks like text classification and information retrieval, where precision and recall are important.
      ii. Other common NLP metrics include:
         1. BLEU: Bilingual Evaluation Understudy metric
         2. METEOR: Metric for Evaluation of Translation with Explicit Ordering
         3. ROUGE: Recall-Oriented Understudy for Gisting Evaluation
2. **Manually evaluate the performance of a model:**
   a. Evaluate multiple prompts with manual evaluations:
      i. The chat playground is an easy way to get started. When you want to manually evaluate multiple prompts more quickly, you can use the manual evaluations feature. This feature allows you to upload a dataset with multiple questions, and optionally add an expected response, to evaluate the model's performance on a larger test dataset.
      ii.


      iii.


      iv. You can rate the model's responses with the thumbs up or down feature. Based on the overall rating, you can try to improve your model by changing input prompt, the system message, the model, or the model's parameters.
3. **Evaluation data:**
   a. To evaluate a model, you need a dataset of prompts and responses (and optionally, expected responses as "ground truth"). You can compile this dataset manually or use the output from an existing application; but a useful way to get started is to use an AI model to generate a set of prompts and responses related to a specific subject. You can then edit the generated prompts and responses to reflect your desired output, and use them as ground truth to evaluate the responses from another model.
   b.
   c.


4. Module Assesment:
   a.


b.


Model benchmarks
Manual evaluations
Correct
Automated evaluations
2. Which evaluator compares generated responses to ground truth based on standard metrics?
Coherence
F1 Score
Correct
Protected material
3. Which evaluator metric uses an AI model to judge the structure and logical flow of ideas in a response?
Coherence
Correct
F1 Score
protected material

c.